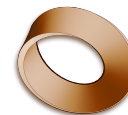


## L3 – The Law of Reflection

Unit 1: Light & EM Spectrum • MYP Sciences Year 8



Physics 8B and 8A • Mr. Rawson

|                  |                                                                                                                                                                                       |               |                                |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------------|
| <b>8B</b>        | Fri 4 Sep, P2, 09:55–10:45                                                                                                                                                            | <b>Next</b>   | Fri 11 Sep, P2 — <b>7 days</b> |
| <b>8A</b>        | Fri 4 Sep, P4, 12:15–13:05                                                                                                                                                            | <b>Next</b>   | Wed 9 Sep, P6 — <b>5 days</b>  |
| <b>Criterion</b> | A — see the written-task note                                                                                                                                                         | <b>Lesson</b> | 3 of $\approx 12$              |
| <b>To print</b>  | <b>32</b> handouts (15 for 8B + 17 for 8A), two pages each. Nothing to cut this time.                                                                                                 |               |                                |
| <b>Kit</b>       | Per group: ray box with single-slit mask, power supply, plane mirror, protractor, plain A4 paper, sharp pencil, 30 cm ruler. <b>Ordered from Joel; not confirmed</b> — red box below. |               |                                |
| <b>Homework</b>  | None built. [[ALEX: decide — whether to set any homework from this lesson]]                                                                                                           |               |                                |

### Kit unconfirmed — check with Joel before Friday

The ray boxes are **ordered from Joel, not confirmed delivered**. Everything below assumes they are in PH1 by P2 Friday, with power supplies and single-slit masks. Group size follows directly from the count: 15 students needs 5 boxes for threes, 8 for pairs; 17 needs 6 or 9.

[[ALEX: decide — how many ray boxes actually arrived, and the group size that follows. And decide the fallback if they have not: the mirror-and-protractor half of the method works with a sighted pencil-and-ruler ray instead of a light ray, but that is your call, not this document's.]]

By the end of the lesson students can:

- tell specular reflection from diffuse reflection, and connect each to the surface that produces it — mirror versus paper;
- draw a ray diagram of reflection with the mirror, the **normal** (dashed), the incident ray and the reflected ray, and label the angle of incidence  $i$  and the angle of reflection  $r$ ;
- state the law of reflection —  $i = r$ , **both measured from the normal** — having found it in their own measurements.

**The written-task question.** The handout's three command-term questions (State / Describe / Explain, definitions printed per the house style) are written evidence this lesson produces — the first chance since the L2 red box flagged that the Clifton Hall rebuild left the unit with no written Criterion A task, and 8A still has none this year. [[ALEX: decide — whether these three questions count as assessed Criterion A evidence or stay formative practice]]

**Timing — run for 8B at P2; 8A deltas below**

| Time  | Phase                       | What happens                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0–5   | Do Now                      | Retrieval from L2, on the board: put infrared, X-rays and radio in order of energy; which band do your eyes detect? Then the hook, asked and left hanging: <i>you can see your face in a mirror — why not in a sheet of paper?</i> Both are reflecting light at you.                                                                                                                                                                                                                           |
| 5–14  | Two kinds of reflection     | Specular versus diffuse, board diagrams: parallel rays stay parallel off a smooth surface, scatter off a rough one. Then the vocabulary of the diagram: incident ray, reflected ray, and the <b>normal</b> — the dashed construction line at $90^\circ$ to the mirror. Say explicitly, twice: <b>both angles are measured from the normal, not from the mirror.</b> Establish the ray model in passing — light drawn as straight lines with arrows; the ray box is about to make that visible. |
| 14–20 | Demo the setup              | Build one station on the front bench, narrating: mirror on the paper line, normal drawn first, ray aimed at the point where the normal meets the mirror. Measure one pair of angles badly on purpose — from the mirror surface — get a non-sense reading, and let them catch it. Safety: bulbs get hot, leads tidy, no shining at eyes; lights down at the back.                                                                                                                               |
| 20–37 | Practical                   | Groups measure and tabulate the five set angles ( $20^\circ$ – $60^\circ$ ), one spare row for a sixth of their choosing. Circulate hunting for one specific error: <b>the protractor zeroed on the mirror instead of the normal</b> — it shows up as a pair summing to $90^\circ$ . Send anyone with $i \neq r$ by more than a couple of degrees back to remeasure, not to shrug.                                                                                                             |
| 37–45 | Diagram + written questions | Kit off and to the side. Handout page 2: the ray diagram on the printed mirror line — normal dashed, rays ruled, arrows on — then the three written questions. The Explain question closes the Do Now hook: mirror versus paper, in the words specular and diffuse.                                                                                                                                                                                                                            |
| 45–50 | Debrief + pack              | Say the law out loud and have them say the qualifier back: <i>measured from where?</i> Close the loop on diffuse reflection: paper obeys the same law at every point — millions of tiny mirrors at different angles. Kit counted back before anyone leaves.                                                                                                                                                                                                                                    |

**8A deltas (P4). 8B runs first this time** — the reverse of Monday — so whatever the practical actually took at P2, re-cut 20–37 before 12:15. 8A is the larger set (17, so one more group); kit stays out between the periods if P3 allows. 8B has done the Lotto write-up and met the three command terms formally; 8A has not, so at P4 read each command-term definition out loud when the written questions start rather than assuming them.

## Questions to ask

1. Why do we measure from the normal and not from the mirror? (*The mirror might be curved — the normal is defined at a single point, so the law works for any surface. For Y8: “so everyone measures from the same line, even on a curved mirror.”*)
2. Your two angles add up to  $90^\circ$  — what have you actually measured? (*One angle from the normal and one from the mirror. The catch question for the main error.*)
3. A sheet of paper reflects light at you — why is there no image in it? (*Diffuse: the surface is rough, so parallel rays leave in all directions and the picture is scrambled.*)
4. Does rough paper break the law of reflection? (*No — every point obeys it; the normals point in different directions.*)
5. If I tilt the mirror a little, what happens to the reflected ray? (*It swings by twice the tilt — the extension question, worth asking only of groups that finish early.*)

## Common misconceptions

- **Measuring from the mirror, not the normal.** The error this lesson exists to kill. Signature: readings that pair to  $90^\circ$  ( $20^\circ$  with  $70^\circ$ ). Circulate for it; the demo stages it deliberately so the class has already seen it fail.
- **The normal treated as a real ray of light.** It is a construction line — dashed on every diagram for exactly that reason. Insist on the dashing.
- **Protractor misread from the wrong scale.**  $40^\circ$  read as  $140^\circ$ . A reflected angle over  $90^\circ$  is the tell.
- **“Rough surfaces don’t reflect” / “don’t obey the law.”** They do both — diffuse reflection is the law applied to millions of tiny surfaces. If this were false you could not see the page.
- **Rays drawn without arrows, or not ruled.** Habits worth enforcing now, cheaply, before the B/C summative marks them.

## Differentiation

**Support:** the handout’s method draws the normal *first*, before the ray box is switched on — a group that is lost gets walked to step 2, not given the answer. The five angles are set, so there is no design load, only measuring.

**Extend:** the spare table row, then the handout’s extension: predict what happens to the angle *between* the two rays when  $i$  goes up by  $10^\circ$  — and then to the reflected ray when the mirror tilts. Quick groups can test both with the kit in front of them.

## After the lesson

- **8B runs first.** Re-cut the practical timing before P4, and note which error actually dominated — it steers the L4 Do Now.
- Flip both Class Log entries to **Taught**; log how many groups the kit allowed, for next time it is ordered.
- This practical is the **on-ramp to the B/C reflection/refraction summative** (unit map). If tabulating and concluding were shaky, that summative needs a scaffolding lesson first — better known now.
- The Criterion A decision (page 1) is still open: if the three written questions counted, mark them; if not, the unit still owes a written task.
- Kit back to Joel or stored per his instruction — first practical of the year, so the routine set here is the routine kept.